Nitya Dhagat

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Professional Summary

IEEE SCEECS 2026 Best Paper Award recipient and author of two preprints, with research spanning ML-based satellite remote sensing, geographic generalization of flood-mapping models, multilevel NLP classification, and LLM-powered conversational systems. MS Data Science candidate (enrolled full-time) at UC San Diego and former researcher at ISRO's Space Applications Centre, focused on solving real-world, large-scale scientific problems.

Education

University of California San Diego — La Jolla, CA
Master of Science in Data Science — *Anticipated: 06/28 (enrolled full-time)*

Dharmsinh Desai University — Nadiad, Gujarat, India
Bachelor of Technology in Information Technology — *Graduated: 05/26*
GPA: 9.35/10 (Top 3 in Department)

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Artificial Intelligence, Probability & Statistics, Data Mining, Distributed Computing, Big Data Analytics

Publications

- N. D. Dhagat, N. P. Desai, "HiDeNET: Hybrid Deep Neural Architecture for Multilevel Sentiment Classification of Cryptocurrency Comments," *IEEE SCEECS 2026* — **Best Paper Award, CSE Track**. ieeexplore.ieee.org/document/11429958
- N. D. Dhagat, "Evaluating Cross-Country Generalization of Deep Learning-Based Flood Mapping Using Sentinel-1 SAR and Sentinel-2 Imagery," *EarthArXiv* preprint, 2026 (sole author). doi.org/10.31223/X59Z2M
- N. D. Dhagat, V. K. Dabhi, H. B. Prajapati, Z. J. Barad, "Telephony Voice Agent for Banking Services," *arXiv*, 2026. arxiv.org/abs/2606.28779
- N. D. Dhagat, N. Agarwal, A. K. Mandal, "A River-Aware Machine Learning Framework for Correcting SMAP Sea Surface Salinity in the Bay of Bengal," *manuscript in preparation*, 2026.

Research Experience

Independent Researcher | Cross-Country Generalization of DL-Based Flood Mapping — May 2026 – Sep 2026
Tech: Python, PyTorch, segmentation-models-pytorch, U-Net, ResNet34, Sentinel-1 SAR, Sentinel-2, Sen1Floods11

- Designed a geographically disjoint country-level evaluation protocol for Sen1Floods11 (train on 7 countries, hold out Paraguay and Somalia); benchmarked 12 configurations across Vanilla U-Net and ResNet34 U-Net with Sentinel-1 SAR, Sentinel-2 + NDWI/MNDWI, and multimodal fusion.
- Showed random-split evaluation overstates real-world performance: Vanilla U-Net + Sentinel-2 IoU fell from 0.727 to 0.464 (36.2% relative loss) under country-level splitting; ResNet34 + Fusion was strongest (IoU 0.555, AP 0.850), narrowing the Somalia-vs-Paraguay gap from 2.4× (SAR) to 1.3×.

Researcher — Dec 2025 – May 2026
Space Applications Centre (SAC), ISRO | Physical Oceanography Division — India
Tech: Python, LightGBM, XGBoost, Random Forest, SVR, NumPy, Pandas, SciPy, cKDTree, NetCDF4/HDF5, SHAP, GloFAS

- Developed a river-aware ML correction framework for SMAP sea surface salinity under Dr. Neeraj Agarwal (Division Head) and Sri. Anup Kumar Mandal; journal manuscript in preparation.
- Built a 1 TB+ pipeline ingesting NASA SMAP L2C RSS V6.0 swath data, Coriolis Argo profiles, and GloFAS river discharge for the Bay of Bengal, with spatial-temporal collocation of SMAP pixels to Argo profiles via k-d tree search on ECEF coordinates.
- Engineered a novel river-discharge feature set (500 km spatial influence, 7-/14-day lagged terms), the first known use of river discharge as a direct ML predictor for SMAP SSS correction.
- Benchmarked LightGBM, XGBoost, Random Forest, and SVR with a strict temporal split (2015–2022 train / 2023 test); XGBoost reduced SSS RMSE from 0.929 to 0.547 PSU (41.2%) and fresh bias from −0.211 to +0.093 PSU, with a 47.1% RMSE reduction in the coastal shelf zone (0.745 to 0.394 PSU).
- Applied SHAP analysis: lagged, spatially aggregated discharge features outranked instantaneous discharge, consistent with 7–14 day coastal-current advection.

Student Researcher | Telephony Voice Agent for Banking Services — May 2025 – Apr 2026
Dharmsinh Desai University — Nadiad, India
Tech: Python, Dialogflow CX, Twilio, Google Cloud Run, Docker

- With Dr. Vipul K. Dabhi, Dr. Harshadkumar B. Prajapati, and Zankhana J. Barad, designed a secure voice-based banking agent using Dialogflow CX and Twilio; achieved 95% accuracy handling interruptions and contextual shifts in real-time conversations; published as arXiv preprint.
- Deployed cloud-native microservices on Google Cloud Run with multi-factor authentication, supporting 99.9% uptime and

concurrent session scalability.

Student Researcher | HiDeNET: Hybrid Deep Neural Architecture for Sentiment Classification May 2024 – Feb 2026

Dharmsinh Desai University

Tech: Python, DeBERTa-v3, 1D CNN, PyTorch, scikit-learn

- With Prof. Nikita P. Desai, designed HiDeNET combining DeBERTa-v3 (global context) and 1D CNN (local features) for multilevel cryptocurrency sentiment classification; 17% improvement over the SVM baseline; accepted at IEEE SCEECS 2026 (Best Paper Award, CSE Track).
- Applied stratified sampling and focal loss to address class imbalance, reducing Hamming Loss by 20%; achieved 80%, 90%, and 87% accuracy across Level 1, 2, and 3 of the sentiment hierarchy.

Professional Experience

Software Engineer Intern Dec 2024 – Apr 2025

Atyantech Private Limited Nadiad (Hybrid)

Tech: Android, Arduino, ESP32, Wi-Fi, JSON, TCP/IP, IoT

- Built a custom TCP server on ESP32 with JSON parsing, improving data transmission efficiency by 30% and processing speed by 50%.
- Engineered a checksum-based error detection system, reducing data corruption by 35% and improving throughput by 40%.

Projects

Twilio-Powered Real-Time Voice Conversational AI | GitHub Link

Tech: Python, Flask, Twilio, Vosk STT, Hugging Face APIs

- Built a modular voice agent that transcribes live phone calls with Vosk STT (sub-300ms latency per utterance, 95%+ accuracy in controlled tests), with two-way audio via Twilio.
- Integrated lightweight LLMs (Phi-2, TinyLlama) and Mistral-7B via the Hugging Face Inference API as an automated fallback to language generation when structured data is not applicable.

MealMind: AI-Powered Food Recommendation System | GitHub Link

Tech: Python (Flask), Scikit-learn, Zomato API, Google Maps API, Heroku

- Developing a recommendation system for real-time, personalized meal suggestions based on preferences, dietary restrictions, and location, integrating Zomato, Google Maps, and OpenWeather APIs for context-aware results.

BlogNest: JWT-Secured Blog Management Platform | GitHub Link

Tech: React, Spring Boot, MySQL, JWT, Hibernate, REST APIs

- Built and deployed a full-stack blog platform (Spring Boot, 24+ REST APIs, MySQL) with 5 core entity models and JWT-based authentication with role-based access control.
- Achieved 40% faster API response times and 60% higher average user engagement through streamlined CRUD operations and efficient backend structuring.

HiDeNET Inference Optimization: Latency and Throughput Study | GitHub Link

Tech: Python, PyTorch, torch.compile, ONNX Runtime, torch.profiler, Hugging Face Transformers, A100 GPU

- Benchmarked `torch.compile`, fp16 autocast, and ONNX Runtime on a PyTorch re-implementation of HiDeNET (within 1.5 points of the paper's accuracy; original checkpoint unavailable): cut A100 batch-1 latency by 60% (22.3 to 8.8 ms), raised batch-16 throughput 2.03× (231 to 468 samples/s), and cut CPU latency 17%, all with 100% prediction agreement against FP32.
- Diagnosed why dynamic INT8 quantization collapsed Level 1 agreement to 29–42% using layer-wise ablations (error concentrated in FFN output layers and accumulating over depth); profiled with `torch.profiler` and showed that trimming padding is not lossless (92.8% agreement) because the Conv1D pooling sees padded positions.

Skills

ML / DL / NLP / CV: PyTorch, TensorFlow, Keras, DeBERTa-v3, Hugging Face, SHAP, LightGBM, XGBoost, scikit-learn, segmentation-models-pytorch, ONNX Runtime, torch.compile, LangChain, Google ADK

Languages & Databases: Python, C++, SQL, Java, JavaScript, C; MySQL, PostgreSQL, MongoDB, Firebase

Data / Scientific Computing: Pandas, NumPy, SciPy, NetCDF4/HDF5, cKDTree, Sentinel-1/2, SMAP, Hadoop, ETL, Jupyter

Cloud / Systems: GCP (Cloud Run), AWS, Docker, Kubernetes, Microservices, GitHub Actions, Prometheus, Grafana

Awards & Leadership

- **Best Paper Award (CSE Track), IEEE SCEECS 2026** (Feb. 2026, MANIT Bhopal): 900+ submissions, <30% acceptance rate; also acknowledged by the Department of IT, DDU.
- **FIRE2024 CryptoQA Track** (Apr. 2024): Ranked 5th (Twitter) and 6th (Reddit) in multilevel crypto sentiment analysis; 63%+ accuracy.
- **Core Committee Member, IETE Student's Forum** (Dec 2023 – May 2026): Ran 10+ seminars (50+ attendees).